

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

05

10 questions · ~15 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

Isabel grows potatoes in her garden. This year, she harvested 760 potatoes and saved 10% of them to plant next year. How many of the harvested potatoes did Isabel save to plant next year?

- A. 66
- B. 76
- C. 84
- D. 86

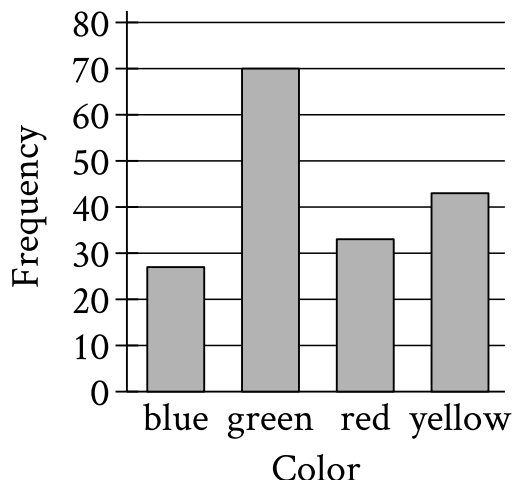
Each face of a fair 14-sided die is labeled with a number from 1 through 14, with a different number appearing on each face. If the die is rolled one time, what is the probability of rolling a 2?

A. $\frac{1}{14}$

B. $\frac{2}{14}$

C. $\frac{12}{14}$

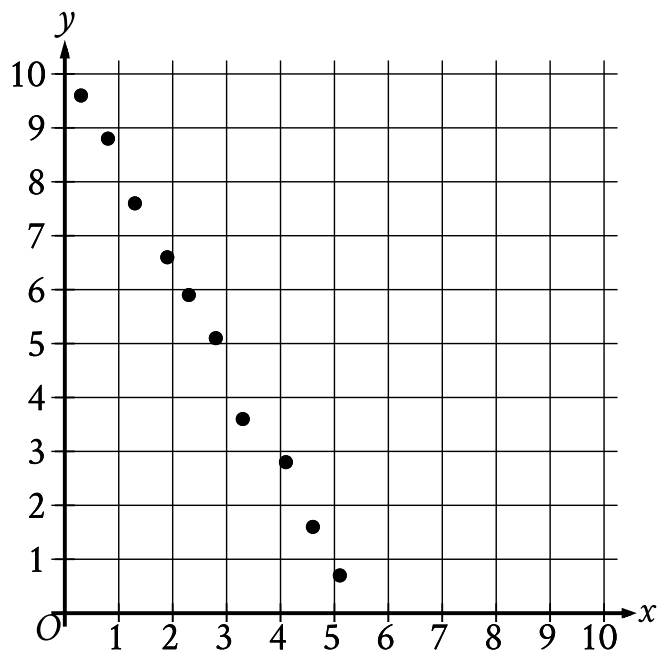
D. $\frac{13}{14}$



- The Frequency data for the 4 bars are as follows:
 - blue: 27
 - green: 70
 - red: 33
 - yellow: 43

A data set consists of 173 colors. The bar graph shows the number of times each color appears in the data set. Which color appears 70 times?

- A. Blue
- B. Green
- C. Red
- D. Yellow



- The scatterplot has 10 data points.
- The data points are in a linear pattern trending down from left to right.
- Select data points have the following approximate coordinates:
 - (0.8 comma 8.9)
 - (2.8 comma 5.1)
 - (4.1 comma 2.9)

Which of the following equations is the most appropriate linear model for the data shown in the scatterplot?

- A. $y = -1.9x - 10.1$
- B. $y = -1.9x + 10.1$
- C. $y = 1.9x - 10.1$

D. $y = 1.9x + 10.1$

A certain bird species can fly at an average speed of 16 meters per second when in continuous flight. At this rate, how many meters would this bird species fly in 4 seconds?

- A. 64
- B. 20
- C. 16
- D. 12

A certain forest is 253 acres. To estimate the number of trees in the forest, a ranger randomly selects 5 different 1-acre parcels in the forest and determines the number of trees in each parcel. The numbers of trees in the sample acres are 51, 59, 45, 52, and 73. Based on the mean of the sample, which of the following ranges contains the best estimate for the number of trees in the entire forest?

- A. 11,000 to 12,000
- B. 12,500 to 13,500
- C. 13,500 to 14,500
- D. 18,000 to 19,000

The length of the base of a certain parallelogram is 89% of the height of the parallelogram. Which expression represents the length of the base of the parallelogram, where h is the height of the parallelogram?

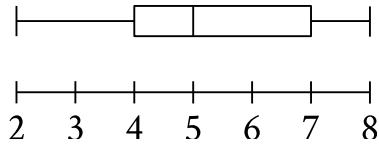
- A. $89h$
- B. $0.089h$
- C. $8.9h$
- D. $0.89h$

Each rock in a collection of 70 rocks was classified as either igneous, metamorphic, or sedimentary, as shown in the frequency table.

Classification	Frequency
igneous	10
metamorphic	33
sedimentary	27

If one of these rocks is selected at random, what is the probability of selecting a rock that is igneous?

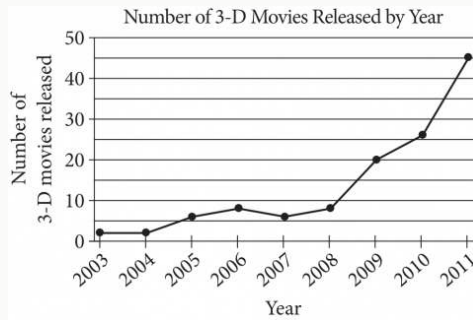
- A. $\frac{10}{27}$
- B. $\frac{10}{33}$
- C. $\frac{10}{60}$
- D. $\frac{10}{70}$



- From left to right the values of the vertical bars in the box plot are as follows:
 - First vertical bar: 2
 - Second vertical bar: 4
 - Third vertical bar: 5
 - Fourth vertical bar: 7
 - Fifth vertical bar: 8

The box plot summarizes 15 data values. What is the median of this data set?

- A. 2
- B. 3
- C. 5
- D. 8



According to the line graph above, between which two consecutive years was there the greatest change in the number of 3-D movies released?

- A. 2003–2004
- B. 2008–2009
- C. 2009–2010
- D. 2010–2011